Artificial Intelligence

Text Book : Peater Norvig
Author : Stwart J. Russell

Unit I : Introduction
Part I : what is AI, foundations of AI, history of AI, the state of Art.
Part II : Intelligent Agents
Agents & environments, good behaviour, the concept of rationality, the nature of environments, The structure of Agents.
Unit II : Solving problems By searching
Problem solving Agents, searching for solutions, on informed search strategies, informed search strategies, Heuristic functions, Beyond classical search : local search algorithms & optimization pobblems, local search in Continuous spaces, searching with non-deterministic actions, searching with partial observations, online search Agents & unknown environments.
Unit III : knowledge representation & Planning
knowledge Based Agents & the wumpus world problem, Propositional logic : Syntax, symantics & theorem proving, Agents Based on propositional problem & model checking, first-order logic : syntax, symantics & inference mechanisms, unification, forword & back wark chaining, resolution, knowledge engineering & onto logical representation, introduction to classical planning & planning algorithms

Unit IV : Reinforcement Learning And Natural Language Processing.

Reinforcement Learning : Introduction, Passive Reinforcement learning, Active Reinforcement learning, Generalization in reinforcement learning, Policy search, applications of reinforcement learning.

Natural Language Processing : Language models, Text classification, information retrieval, information extraction.

Unit V : Robotics & Computer Vision.

Robotics : Introduction, Robotic Hardware, Robotic Perception, Planning to move, Planning uncertain movements, moving, Robotics software Architectures, application domains.

Computer Vision : Phylosophical foundation, the limits of AI ; Can machines Really think? the ethics of AI.